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LRI Emergency Department

Bradycardia management

Version 70

Intended for adults with a heart rate of less than 60 beats per minute

Use in conjunction with guidance concerning the management of specific reversible causes, such as STEMI or hyperkalaemia

***Disclaimer*: This is a clinical template; clinicians should always use judgment when managing individual patients**

Re-approved by ED guideline committee on 25Feb26 . Trust Ref: C189/2016
Review date: Feb 2031

Patient details

Full name

DoB

Unit number

(use sticker if available)

To prescribe in NC Meds, go to Emergency Medicine (ED) > Common scenarios (ED) > Bradycardia

Adverse signs, or risk of asystole (see boxes 1&2)?
- N → Manage as appropriate
- Y →

- ABCDE approach in resuscitation room (ER)
- Involve ED middle grade or consultant **NOW**
- Attach to pacing defibrillator as shown in box 3
- IV access; bloods including VBG and TFTs
- Identify and treat any reversible causes **NOW** +
- 12-lead ECG

Glycopyrronium bromide * — one vial of 200 micrograms in 1mL

OR

Atropine * — one vial of 600 micrograms in 1mL (or 2mL from 3mg in-10mL Minijet)

NB: Use glycopyrronium if patient is at risk of delirium. It has the same mechanism of action as atropine but does not cross blood-brain barrier.

Good response? (i.e. reversal of adverse features)
- N → *If adverse features,* repeat glycopyrronium (or atropine) up to 4x while planning next action * — **NB**: This might not work; particularly in complete AV block with broad QRS complexes
- Y →

Asystole risk (see box 2)?
- N → Where indicated, treat underlying cause
- Y →

Persistent adverse features (see box 1) or special circumstances (see box 4)?
- N → Where indicated, treat underlying cause
- Y →

Pacing needed **NOW**? (i.e. patient peri-arrest or unconscious)
- N →
- Y → Start transcutaneous pacing (**TCP**; see box 6) †

Special circumstances (see box 4)?
- N → Is chemical pacing preferred, or is **TCP** failing to capture?
  - N → Start / continue pacing (see box 6) †
  - Y →
    - Give isoprenaline (see box 7)
    - If not stabilising §, consider switching to an adrenaline infusion (see box 8)
    - → Urgent internal pacing needed; see box 5 for how to arrange
- Y → Treat as per boxes 4 & 9; consider ITU referral **EARLY**

Good response?
- N → Start / continue pacing (see box 6) † → Urgent internal pacing needed; see box 5 for how to arrange
- Y → Stop pacing (if started) † → Where indicated, treat underlying cause + For appropriate disposition, see box 5

\+ **Possible causes include**
- **Medications; e.g.**
  - Beta blocker
  - Calcium channel blocker
  - Digoxin
  - Amiodarone
- **Cardiac** (with or without AMI)
  - AV block
  - Sinus node disease
- **Non-cardiac**
  - ***Hyper / hypokalaemia***
  - Hypo / hypercalcemia
  - ***Acute blood loss***
  - ***Hypothermia***
  - Hypothyroidism
  - Anorexia nervosa
  - Vasovagal episode
- **Physiological; e.g.**
  - Athletes
  - During sleep

* ***DO NOT*** *give either in heart transplant recipient*

§ Persistent instability requires urgent discussion with cardiology team; options include rapid transfer to CCU for TVP (transvenous pacing) with or without changing to an adrenaline infusion, or (exceptionally) TVP at the LRI (**NB**: cardiologists will bring their own kit)

† with adequate sedation / analgesia

NB: percussion pacing can help buy time until transcutaneous pacing is established
- With the side of a closed fist deliver repeated firm blows to the precordium lateral to the lower left sternal edge
- Raise hand about 10cm for each blow

Assessment carried out by

Print name	Signature	Role	Date

① Adverse signs present?

☐ **Yes** - at least one of the below
- Shock ☐
- Syncope ☐
- Myocardial ischaemia ☐
- Acute decompensated heart failure ☐

☐ **No** - none of the above

② Asystole risk present?

☐ **Yes** – at least one of the below
- Complete AV block with
  - broad QRS complexes ☐
  - heart rate < 40bpm ☐
- 2^{nd} degree AV block Möbitz type II ☐
- Ventricular pauses > 3sec ☐
- Recent asystole ☐

☐ **No** – none of the above

③ Pacing preparations

Ensure the defibrillator in patient's bay has pacemaker function (circled below).

Centre anterior defibrillator (pacing) pad over 'V3' ECG chest lead position.

Posterior pad should mirror this position.

*Ensure the defibrillator's ECG electrodes are attached also (**NB**: place shoulder electrodes on the back if possible)*

④ Special circumstances?

☐ **YES** - at least one of the below
- Spinal cord injury with neurogenic shock ☐
- Heart transplant recipient ☐
  - ***Aminophylline 100mg IV (give 1mL (25mg) aliquots per minute; give a further 100mg if required)***
- Beta blocker toxicity ☐
  - ***Glucagon 5–10mg IV then start infusion (see box 9 on reverse)***
  - ***NB: If GlucaGen® 1mg powder kit is not available, High-Dose Insulin Euglycaemic Therapy (HIET; see protocol) on ITU required instead***
- Calcium channel blocker toxicity ☐
  - ***Calcium Chloride 10% 10mL IV***
  - ***High-Dose Insulin Euglycaemic Therapy (HIET) on ITU; see above***
- Digoxin toxicity ☐
  - ***Consider giving Digoxin-specific antibody fragments (DigiFab) 2 vials (80mg; in ER fridge)***

***NB**: Consult* toxbase.org in every case of bradycardia induced by the drugs above

☐ **NO** - none of the above

⑤ Appropriate disposition

CCU (unless patient requires critical care):
- Need for urgent internal pacing; to arrange:
  - Bleep CCU 'registrar' on ***88-2584-[extn]**
  - Try CCU on **13774** or **13719** if no answer
  - If unable to locate, call (via switchboard)
    - In hours: CCU consultant
    - OOH: Non-interventional cardiologist
  - Ensure e-referral is completed
  - Check CCU is expecting pt before transfer
- NSTEMI/STEMI
- Risk of asystole if cause is cardiac

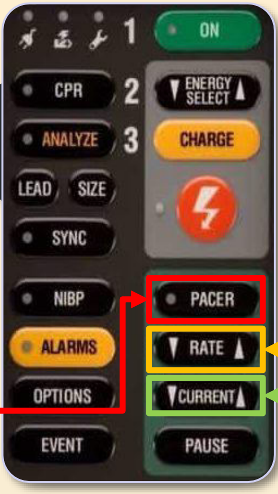
ACB (unless patient requires critical care):
Requiring treatment for bradycardia due to
- Medication side effects or overdoses
- Hypothermia
- Potassium or calcium derangement

Critical care:
- Pt too sick for CCU/ACB (e.g. receiving HIET)
- Hyperkalaemic patients requiring CVVH

CDU:
Patients with cardiac cause of bradycardia not needing transcutaneous or chemical pacing, as deemed suitable by on-call CCU middle grade

In all other patients, use clinical judgment

⑥ Transcutaneous pacing (TCP)



1. Press '**PACER**' button (this will activate the pacing function but not yet start pacing)
2. Press '**RATE ▲**' button (set to 60bpm by default but 70 or 80 bpm may be more effective)
3. Start pacing by pressing '**CURRENT ▲**' button. Repeat until QRS complex is seen after every pacing spike.

   For safety, consider increasing current by a further 10mA.
4. Feel pulse to confirm mechanical capture
5. Ensure good analgesia +/- sedation as needed

⑦ Isoprenaline Hydrochloride

- Add 2mg of isoprenaline hydrochloride (i.e. 2 ampoules of the 1mg in 5mL preparation) to 500mL of glucose 5% (final concentration = 4 microgram/mL)
- Using a 20mL syringe, remove 15mL from this bag to use as bolus
- Give 5mL (20 microgram) IV bolus over 60 seconds
- Repeat bolus up to twice if required
- If bolus worked, start infusion at 5 microgram/min (i.e. set pump to 75mL/h); titrate as needed as per table below

Dose in microgram/min	Infusion rate in mL/h
1	15
2	30
3	45
4	60
5	**75**
6	90
7	105
8	120
9	135
10	150

⑧ Adrenaline

- Add four 1mg/1mL ampoules of adrenaline 1:1000 (= 4mg) to 246mL of sodium chloride 0.9% (final concentration = 16 microgram/mL)
- Start infusion at 3.5 microgram/min (i.e. set pump to 13mL/h)
- Dose range is 2-10 microgram/min; titrate as needed as per table below

Dose in microgram/min	Infusion rate in mL/h
2	7
3	11
3.5	**13**
4	15
5	19
6	22
7	26
8	30
9	33
10	37

⑨ Glucagon

- ***NB*: If GlucaGen® 1mg powder kit not available, DO NOT attempt to give glucagon; ITU team to use HIET**
- Give 5mg IV neat over 1min
- Repeat once if necessary
- Add further 10mg to 40mL of sodium chloride 0.9% in a 50mL syringe (final concentration = 0.2mg/mL)
- If bolus worked, start IV infusion at 0.1mg/kg/h
- See below for the suggested starting rate (dose calculated for a 70kg patient)
- Dose range is 0.05-0.15mg/kg/h up to a maximum rate 10mg/h; titrate as needed
- **Call pharmacist to arrange emergency replenishment as soon as infusion started**

Patient weight in kg	Dose in mg/h	Infusion rate in mL/h
	2	10
	3	15
	4	20
50	5	25
60	6	30
70	**7**	**35**
80	8	40
90	9	45
100	10	50

⑩ Positive chronotrope infusions: How to prescribe on UHL paper chart during NC downtime

PARENTERAL INFUSIONS

	Infusion Fluid		Additions to Infusion				
Date	**Type/Strength**	**Vol.**	**Medicine**	**Dose**	**Route**	**Time to run or ml/hr**	**Prescriber**
DD/MM/YY	*Glucose 5%*	*500mL*	*Isoprenaline hydrochloride*	*2mg = 10mL*	*IV*	*15 - 150mL/h (start at 75mL/h)*	*YOUR NAME & signature*

	Infusion Fluid		Additions to Infusion				
Date	**Type/Strength**	**Vol.**	**Medicine**	**Dose**	**Route**	**Time to run or ml/hr**	**Prescriber**
DD/MM/YY	*Sodium chloride 0.9%*	*246 mL*	*Adrenaline*	*4mg = 4mL of 1:1000*	*IV*	*7 - 37mL/h (start at 13mL/h)*	*YOUR NAME & signature*

	Infusion Fluid		Additions to Infusion				
Date	**Type/Strength**	**Vol.**	**Medicine**	**Dose**	**Route**	**Time to run or ml/hr**	**Prescriber**
DD/MM/YY	*Sodium chloride 0.9%*	*40mL*	*Glucagon*	*10mg = 10 1mL vials*	*IV*	*10 - 50mL/h (start at 35mL/h)*	*YOUR NAME & signature*